Inference of Network Structure from Dynamics in Bayesian Networks

Course: Statystyczna Analiza Danych 2
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Github Repo: `https://github.com/MichalSlowakiewicz/SAD2_project`

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1 Introduction

The purpose of this project is to investigate how the type and amount of data describing network dynamics influence the accuracy of inferring network structure within the framework of Bayesian networks.

This project is divided into two parts:

1. **Systematic Analysis:** We construct random Boolean networks of varying sizes and simulate them under different conditions (synchronous vs. asynchronous updates) to generate trajectories datasets with distinct characteristics (e.g., transient-heavy vs. attractor-heavy).

2. **Biological Application:** We apply the insights gained from the synthetic analysis to reconstruct the structure of a real-life biological process—the regulation of neurogenesis by *miR-9*—using a validated model from the Biodivine repository.

2 Methodology & Experimental Setup

2.1 Network Inference Tool

We utilized **BNFinder2**, a software tool designed for reconstruction of dynamic Bayesian networks (DBNs). Unlike classical Bayesian networks, DBNs allow for the modeling of feedback loops (cycles), which are prevalent in biological systems. The software was executed in a **Python 2** environment as required.

2.2 Scoring Functions

To evaluate potential network structures, we employed two scoring criteria provided by BNFinder2:

- **Minimal Description Length (MDL):** A criterion that balances model accuracy with complexity to prevent overfitting.
- **Bayesian-Dirichlet equivalence (BDe):** A Bayesian scoring metric that assumes uniform prior probabilities over structures.

2.3 Evaluation Metrics

To assess the accuracy of the reconstructed networks against the ground truth, the following metrics were chosen:

Structure-Based Graph Distance Measures:

- **Structural Hamming Distance (SHD):** Represents the minimum number of edge insertions, deletions, or flips required to transform the inferred graph into the ground truth graph.
- **Jaccard Distance:** Measures the dissimilarity between the set of edges in the inferred network and the ground truth.

Classification Metrics:

- **Precision, Recall, and F1 Score:** Calculated based on the count of True Positives (TP), False Positives (FP), and False Negatives (FN) of inferred edges.

3 Part I: Analysis of Synthetic Boolean Networks

3.1 Network Construction

We generated random Boolean networks with sizes $N \in \{5, 7, 10, 16\}$ nodes. The generation process adhered to the following constraints:

- **Topology:** Each node was assigned a maximum of 3 parent nodes.
- **Logic:** Boolean functions for each node were generated at random.

3.2 Data Simulation

For each network, trajectories were simulated in both **Synchronous** and **Asynchronous** update modes. To rigorously test the impact of data characteristics, we generated four distinct types of datasets for every network configuration:

1. **Full Dynamics:** The entire simulated trajectory.
2. **Transient-Heavy:** A dataset consisting primarily of the initial transient states (beginning of simulation).
3. **Attractor-Heavy:** A dataset capturing the steady-state behavior (end of simulation).
4. **Sub-sampled:** The entire trajectory sampled every **3rd time step** to simulate lower temporal resolution.

Additionally, we varied the dataset size by considering:

- **Single Trajectory:** A limited data scenario.
- **Multiple Trajectories:** A rich data scenario consisting of **300 trajectories**.

3.3 Results

We evaluated the results on precision, recall and F1 score, as well as graph specific metrics: Jaccard Distance and SHD.

3.3.1 Impact of Dataset Type and Size

Below is the comparison of different dataset types and its impact on the recreation of original Boolean network. In each graph we compare single and multiple trajectories dataset along with its subsets: Full Dynamics, Transient-Heavy, Attractor-Heavy, and Sub-sampled.

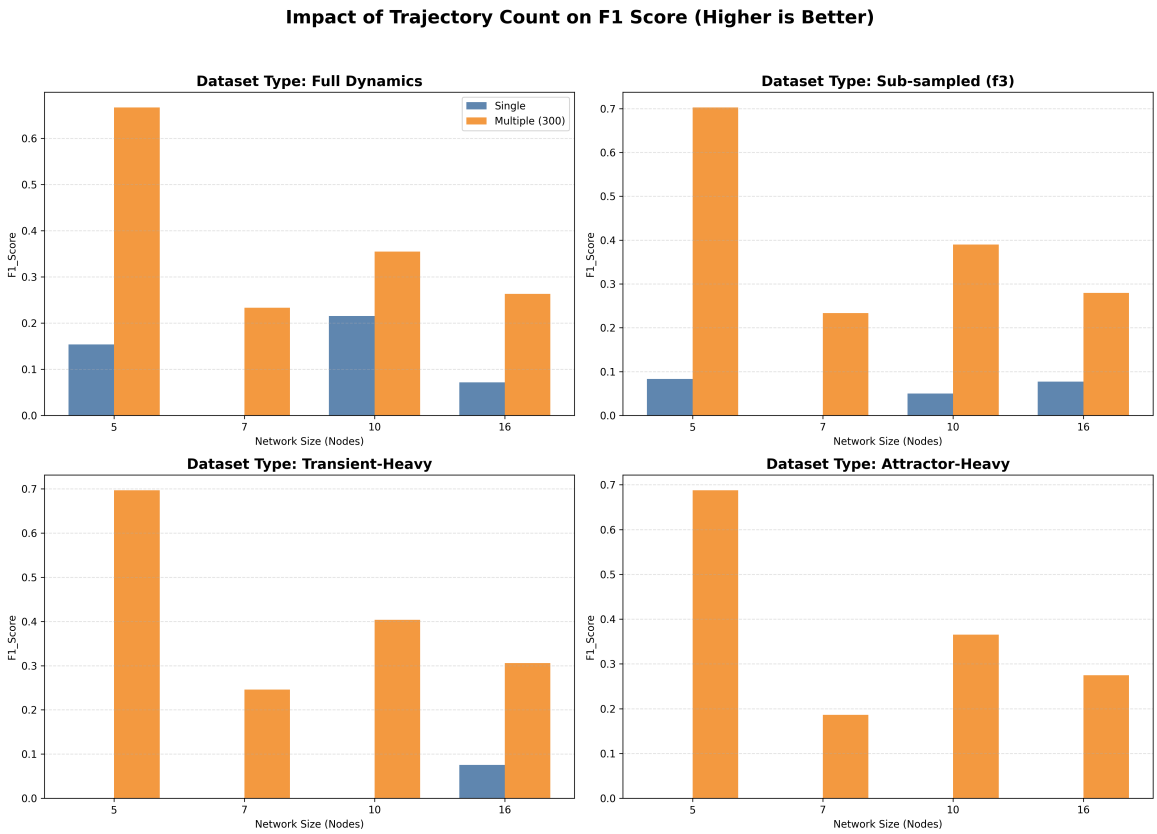

Figure 1: Classic Metrics Comparison - F1 score

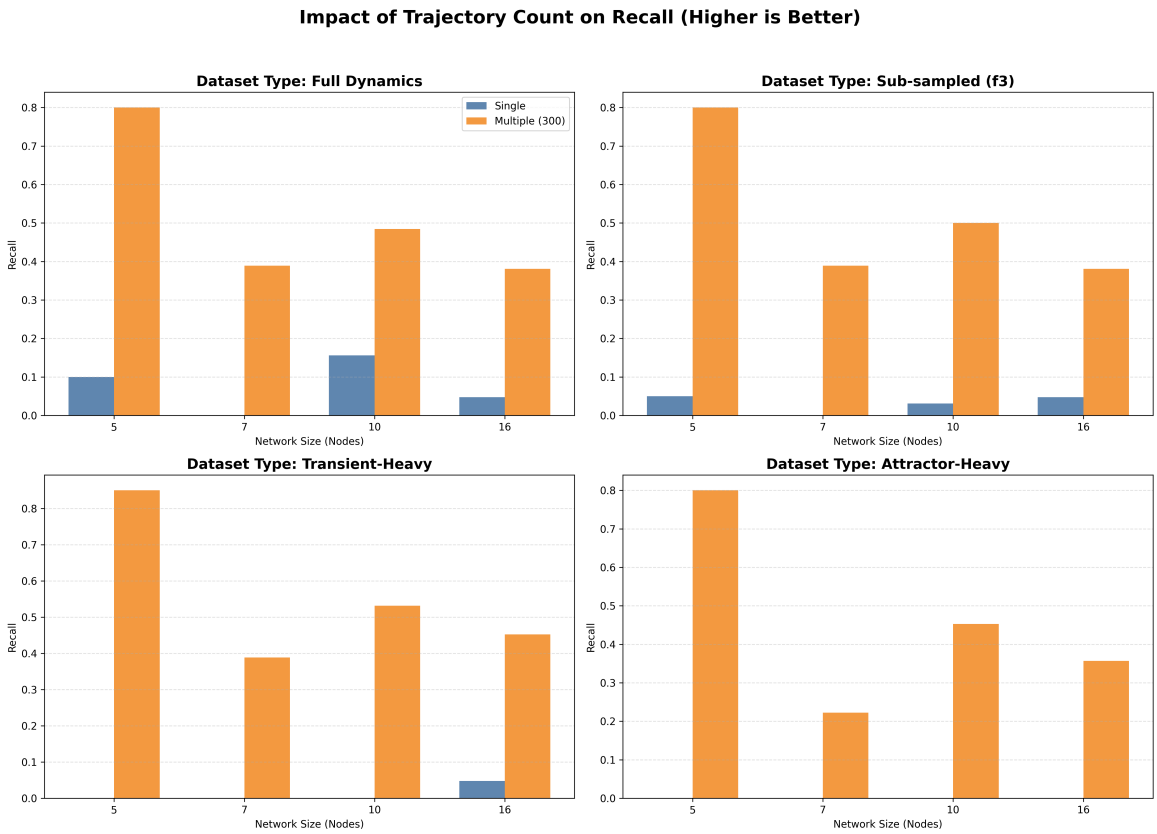

Figure 2: Classic Metrics Comparison - Recall

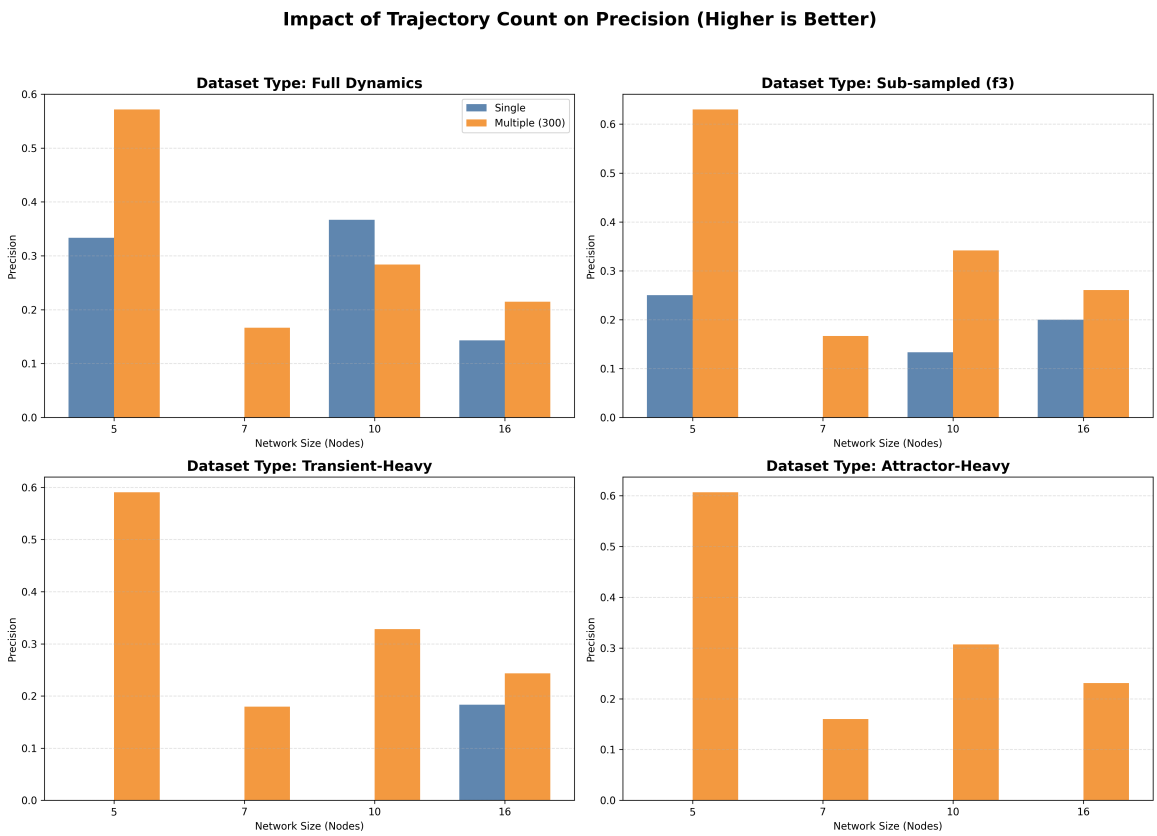

Figure 3: Classic Metrics Comparison - Precision

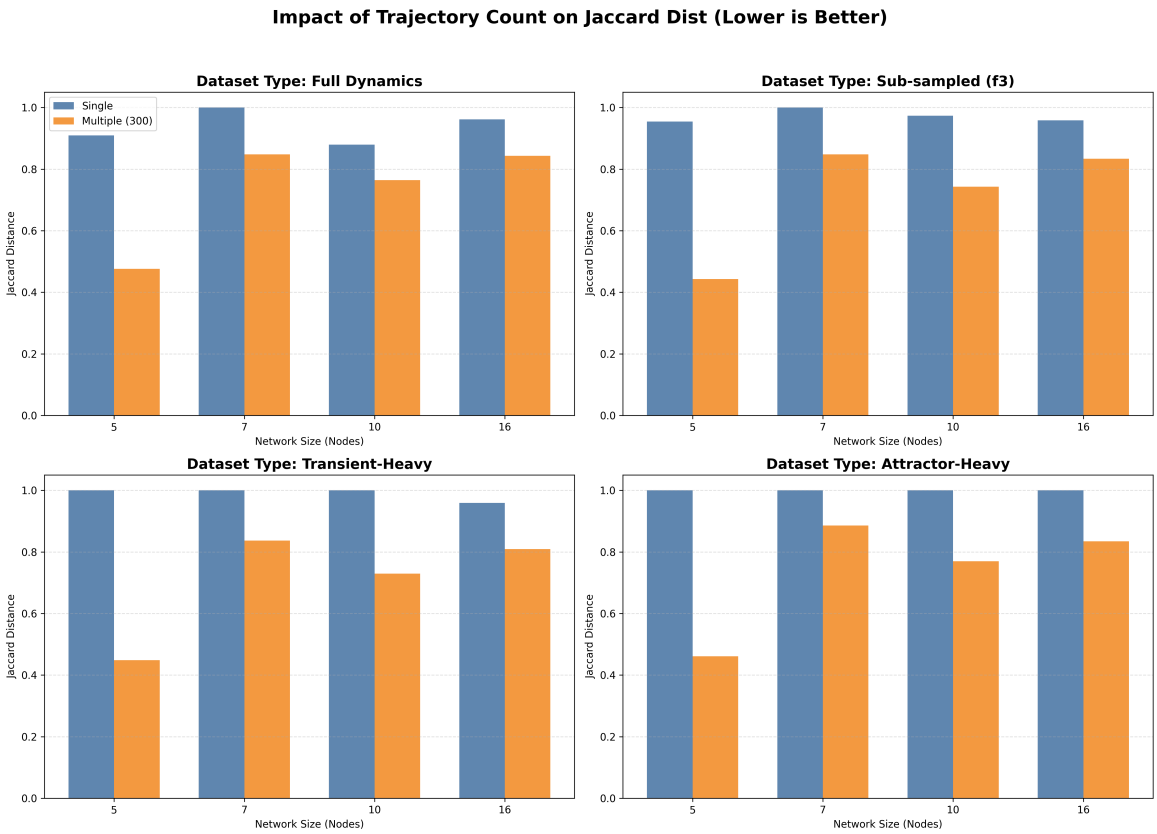

Figure 4: Graph Metrics Comparison - Jaccard Distance

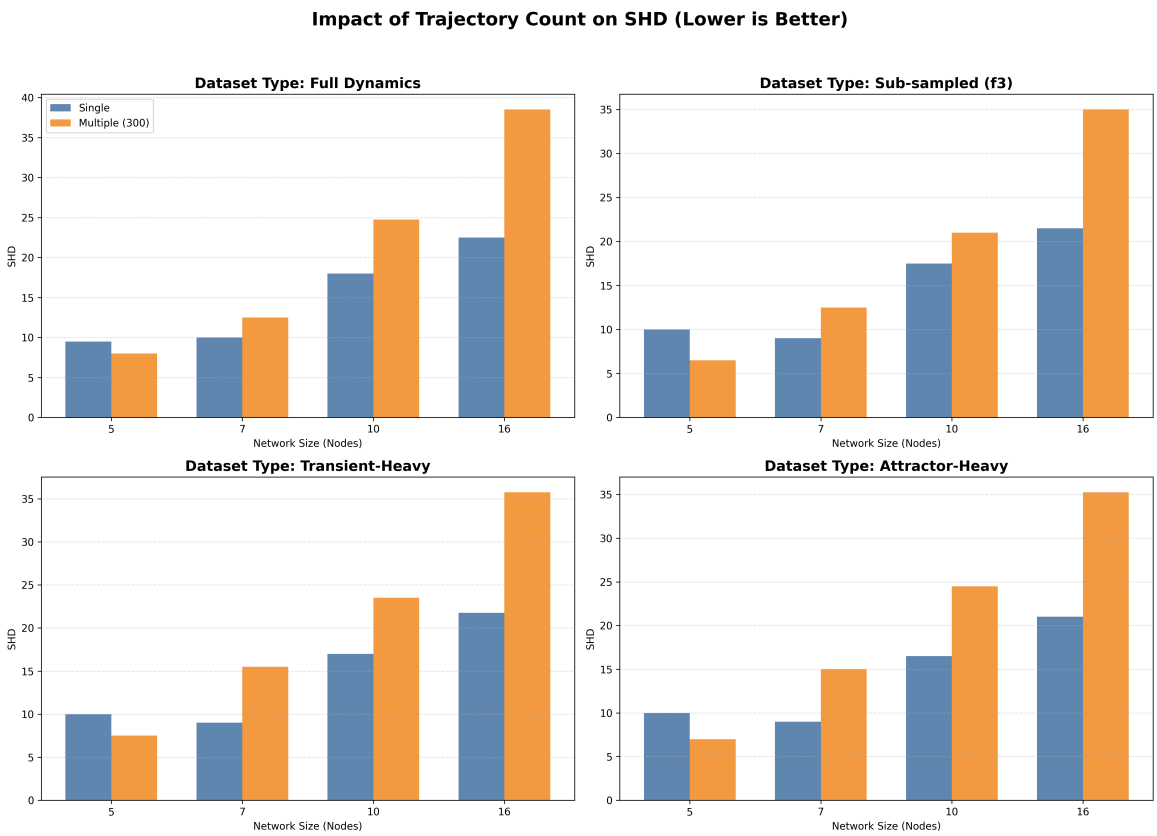

Figure 5: Graph Metrics Comparison - Structural Hamming Distance

Observation: Across all classic metrics, multi trajectories dominated the results. Interestingly, even though Jaccard distance was also better for across all datasets and networks for multiple trajectories, SHD had better results for the smallest network on single trajectory, however, the difference was not significant. Full-sample, attractor-heavy, transient-heavy and sub-sampled datasets performed extremely similar and there was not a clear winner.

3.3.2 Impact of Regime and Scoring

We wanted to verify which regime (Synchronous vs Asynchronous) was better suited for smaller networks, since the network we chose for the second part had 6 nodes. In order to do that, we analyzed F1 scores and both graph metrics for full-dynamics, multiple trajectories datasets, which, according to the results above, were reconstructed with highest results. Additionally, we compared different scoring options of BNFinder (MDE vs MDL).

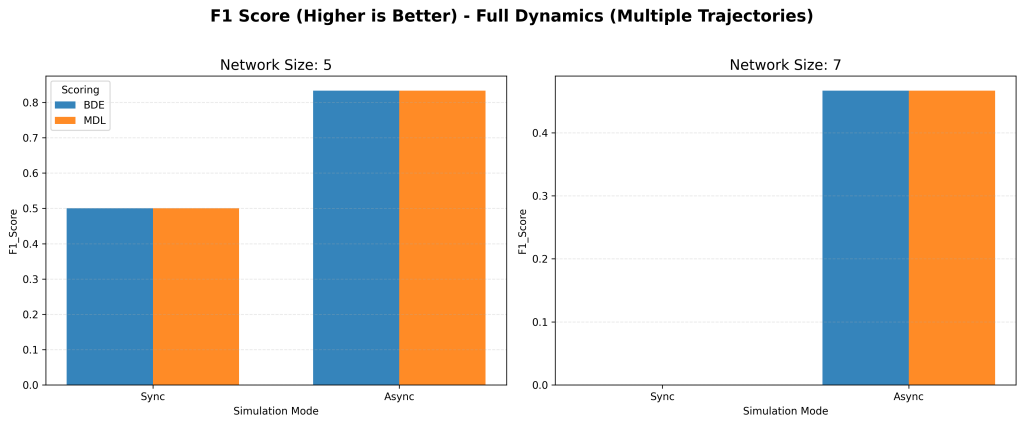

Figure 6: Regime and Scoring Impact - F1 Score

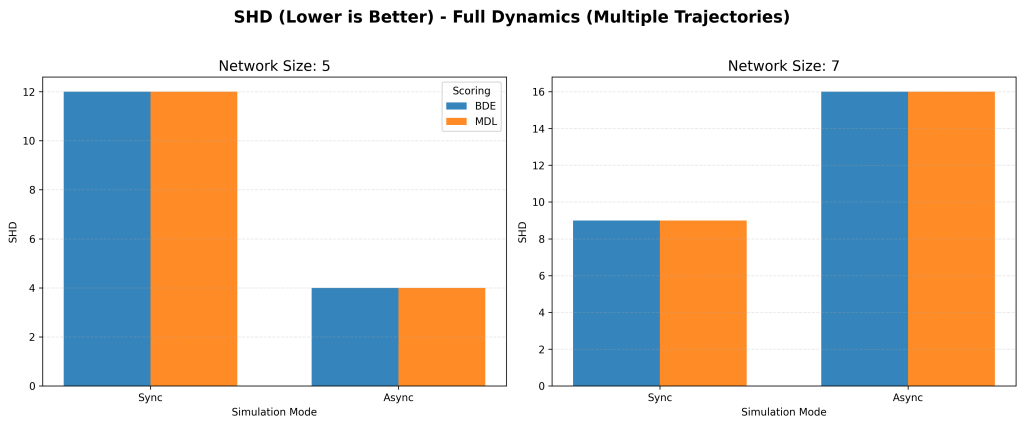

Figure 7: Regime and Scoring Impact - Structural Hamming Distance

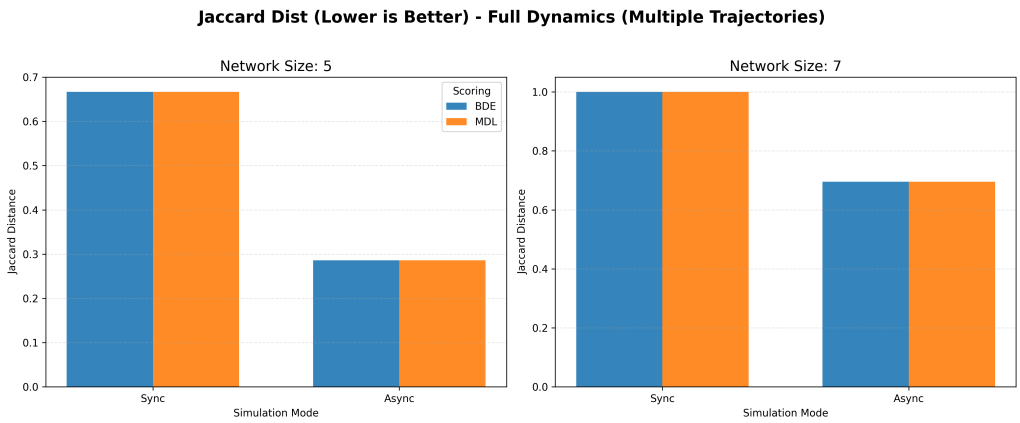

Figure 8: Regime and Scoring Impact - Jaccard Distance

Observation: As visible on the charts above, asynchronous regime outperformed synchronous one across all metrics. The choice of scoring appeared to have no effect on the performance of BNFinder in this cases.

4 Part II: Real-Life Biological Process Reconstruction

4.1 Model Selection

For the real-world application, we selected the **Neurogenesis Regulation by miR-9** model (BBM-088) from the Biodivine repository.

- **Source:** Coolen et al. (2012), "miR-9 controls the timing of neurogenesis through the direct inhibition of antagonistic factors".
- **System Size:** 6 Nodes.
- **Biological Context:** This model describes the gene regulatory network controlling the timing of differentiation in the zebrafish hindbrain.

4.2 Experimental Application

Motivated by results of Part 1, we chose to generate full trajectories in asynchronous regime. Due to the fact that the scoring did not have a meaningful impact on the results, we used both MDL and BDE functions for BNFinder.

4.2.1 Dataset Generation

Based on the insights from Part I, we generated a synthetic dataset for the *miR-9* model to ensure optimal conditions for reconstruction.

- **Simulation Mode:** Asynchronous
- **Dataset Characteristics:** We have generated 4 datasets, with the following characteristics:
  - 101 trajectories of 100 steps
  - 101 trajectories of 200 steps

  - 101 trajectories of 500 steps
  - 101 trajectories of 1000 steps

- **Sampling:** We used full resolution, using all steps from trajectories.

4.2.2 Network Inference Configuration

We then applied BNFinder2 to this generated dataset to infer the network structure.

- **Scoring Function:** We selected both MDE and MDL scoring functions because they performed identically best in our synthetic experiments.
- **Settings:** We configured BNFinder2 to search for Dynamic Bayesian Networks (allowing cycles).

4.3 Evaluation Results

The reconstructed network was compared against the validated Boolean rules from Coolen et al. (2012). We have calculated F1 score, Jaccard Distance and SHD metrics in order to evaluate the final results. Additionally, we compared trajectory lengths and scoring functions to find the best performing configuration for reconstructing the network.

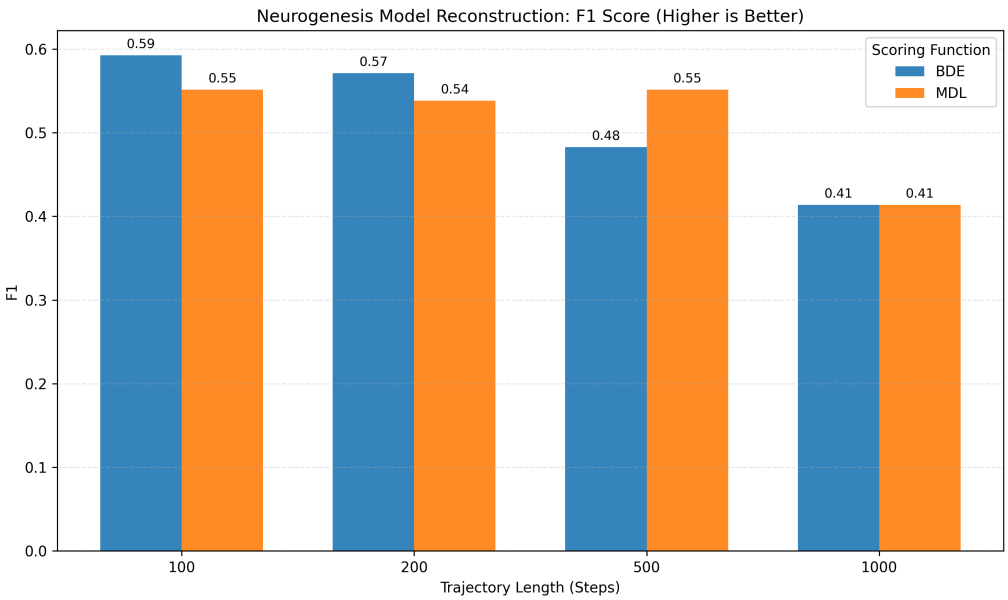

Figure 9: Biological Model - F1 Score Results

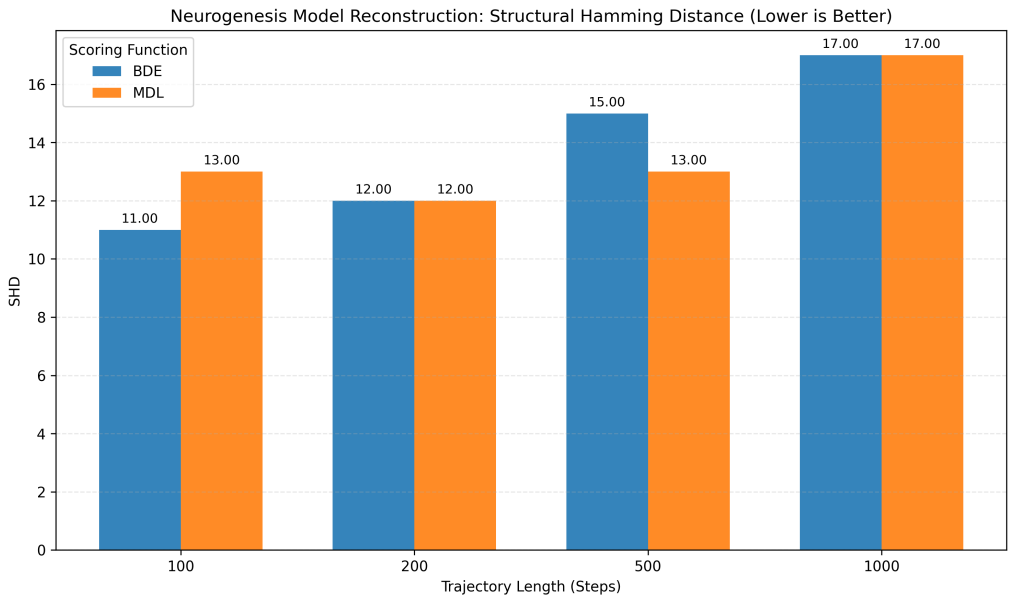

Figure 10: Biological Model - Structural Hamming Distance Results

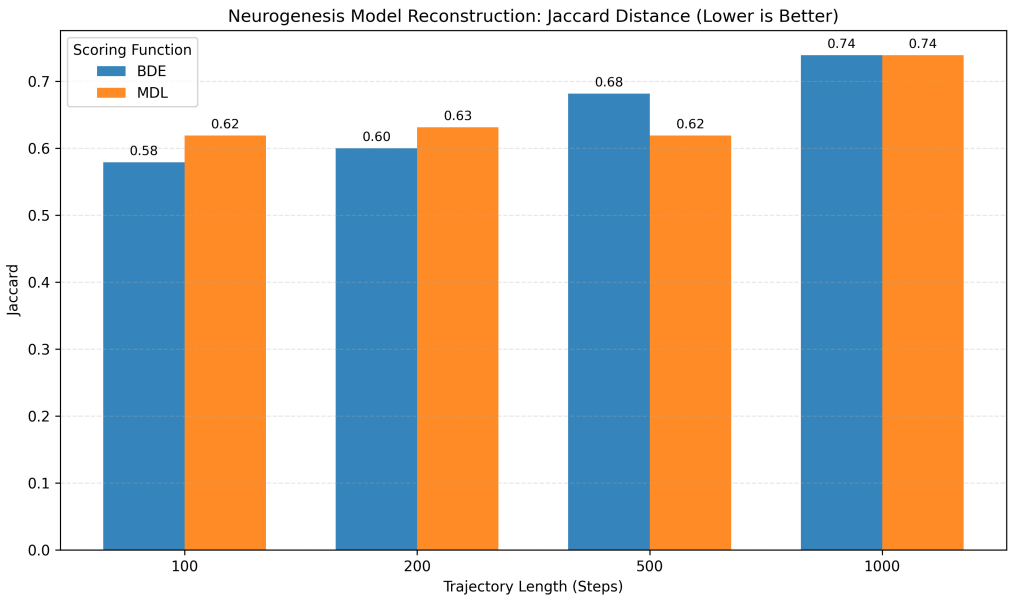

Figure 11: Biological Model - Jaccard Distance Results

Observation: Network reconstruction performed the best across all metrics using BDE scoring function and the smallest trajectory length.

5 Discussion

5.1 Impact of Data Characteristics on Inference Accuracy

Our results demonstrate a clear distinction between single-trajectory and multiple-trajectory datasets. Increasing the dataset size to 300 trajectories generally led to a **significant** improvement in reconstruction accuracy across all metrics, particularly for larger networks (10 and 16 nodes). For instance, in the "Full Dynamics" dataset, the F1 score for the 10-node network jumped from approximately 0.2 (Single) to nearly 0.4 (Multiple).

Furthermore, comparing the different sampling strategies revealed **no significant difference** in inference accuracy across the tested conditions. The F1 scores for "Transient-Heavy" versus "Attractor-Heavy" datasets were highly comparable, indicating that both dynamic phases provided sufficient causal information for reconstruction. Similarly, subsampling the trajectories (every 3rd step) yielded results that were virtually indistinguishable from the "Full Dynamics" datasets. This suggests that the inference process is robust to lower temporal resolution and that the critical causal signals are preserved even when the dataset is reduced or focuses on specific dynamic regimes.

5.2 Scoring Functions: MDL vs. BDe

Throughout the synthetic experiments, the choice of scoring function appeared to have a negligible effect on performance. As seen in the "Full Dynamics" comparison for network sizes 5 and 7, both MDL and BDe produced nearly identical F1 scores and Jaccard distances. However, in the biological reconstruction (Section 4.3), **BDe** slightly outperformed MDL for shorter trajectory lengths (100 and 200 steps), achieving the highest F1 score of 0.59 compared to MDL's 0.55. This suggests that for this specific biological topology, **BDe's likelihood estimation** was slightly more effective at recovering true edges from limited data, although both performed similarly as data volume increased.

5.3 Reconstruction of the Biological Model

For the *miR-9* regulatory network, the reconstruction achieved a maximum F1 score of **0.59** using the BDe scoring function on the 100-step dataset.

- **Successes:** We successfully recovered key interactions defined in the validated model, reflected in the relatively low Structural Hamming Distance (SHD) of 11.00 for the best-performing configuration.
- **Challenges:** The algorithm's performance degraded as trajectory length increased (F1 dropped to 0.41 at 1000 steps). This counter-intuitive result suggests the algorithm struggled to identify **complex regulatory logic** when the dataset became dominated by redundant steady-state samples, leading to overfitting of the attractor dynamics at the expense of the causal logic visible in the transient phase.

5.4 Conclusions

In conclusion, this study confirms that dynamic Bayesian network inference is sensitive to the sampling strategy, but most notably to the regime (synchronous vs asynchronous). To maximize accuracy in real-world scenarios, experimental design should prioritize **capturing dynamics** over merely increasing the duration of observation. Short, dense time-series data captured during the system's response to perturbation appears to yield richer causal information than long-term steady-state monitoring.

6 Contribution Statement

- **[Michał Słowakiewicz], Part I of the project:**
  - Network Generation: Designed and implemented a Python pipeline to generate random Boolean Networks with varying sizes and restricted connectivity along with the "ground truth" for each network.

- Simulation Engine: Developed a robust simulation engine for both synchronous and asynchronous update modes.
- Inference Automation: Constructed the pipeline for the learning process using BNFinder2 with both MDL and BDE scoring criteria across all generated datasets
- Performance Evaluation: Implemented an evaluation module comparing inferred networks against the ground truth. This included calculating classification metrics (Precision, Recall, F1-Score) and structure-based graph distances (Structural Hamming Distance, Jaccard Distance).

- **[Jakub Woźniak], Part II of the project:**
  - Custom Data Generation Pipeline: Developed a specialized data generation pipeline based on the theoretical description of the real model. This involved translating biological descriptions into a computational model and generating the precise "Ground Truth" structure for validation.
  - Simulation Strategy: Designed the simulation methodology, including the concept of re-initializing the network multiple times within a single dataset generation process. This ensured diverse state-space coverage, allowing the BNFinder algorithm to learn from varied biological contexts.
  - Evaluation Script Adaptation: Refactored and optimized the evaluation scripts from Part I to analyze the real biological model. This required adjusting the parsing logic to handle specific ground truth formats (logical equations) and ensuring compatibility with the inference results.
  - Experimental Design (Sample Size Analysis): Proposed and defined the scope of the sensitivity analysis experiment. Conceived the idea to investigate how the length of the time-series data (number of steps/samples) impacts the stability and accuracy of the network reconstruction metrics.

- **[Piotr Okniński], Report and analysis:**
  - **Synthetic Network Performance Analysis:** Conducted a comprehensive analysis of the Part I results, identifying optimal hyperparameters (scoring functions and trajectory types) to refine the experimental design for the biological application.
  - **Biological Inference Evaluation:** Evaluated the reconstruction accuracy of the *miR-9* regulatory network and developed comparative visualizations of final metrics across varying trajectory lengths.
  - **Computational Optimization & Synthesis:** Performed extensive debugging and code refactoring to integrate independent modules into a unified, reproducible repository for all project tasks.
  - **Technical Reporting:** Authored the final report, synthesizing project specifications, graphical data representations, and the theoretical discussion of dynamic Bayesian network inference.

Together, we created the general outline of the project and held the final discussion of project's results.

7 References

1. Coolen, M., Thieffry, D., Drivenes, Ø., Becker, T. S., & Bally-Cuif, L. (2012). miR-9 controls the timing of neurogenesis through the direct inhibition of antagonistic factors. *Developmental Cell*, 22(5), 1052-1064.

2. *BNFinder2 User Manual & Tutorial*, University of Warsaw. Available at: `https://bioputer.mimuw.edu.pl/software/bnf/`

A Synthetic Network Specifications

Specification (Size 5):
The following Boolean functions defined the ground truth for the 5-node synthetic network used in Part I:

```
x1 = (~x5 & x2 & x3) | (x5 & ~x2 & ~x3) | (x5 & ~x2 & x3) | (x5 & x2 &
    x3)
x2 = (x3)
x3 = (~x1 & ~x3 & ~x4) | (x1 & ~x3 & x4) | (x1 & x3 & ~x4)
x4 = (~x4 & ~x3 & ~x2) | (~x4 & ~x3 & x2) | (x4 & ~x3 & x2) | (x4 & x3
    & x2)
x5 = (x2 & x1)
```

Specification (Size 7):
The following Boolean functions defined the ground truth for the 7-node synthetic network used in Part I:

```
x1 = (~x2 & ~x5 & x7) | (~x2 & x5 & ~x7) | (x2 & x5 & x7)
x2 = 0
x3 = (~x7 & ~x6) | (x7 & ~x6) | (x7 & x6)
x4 = (~x2 & ~x4) | (x2 & x4)
x5 = (~x6 & ~x3) | (x6 & ~x3)
x6 = 1
x7 = 0
```

Specification (Size 10):
The following Boolean functions defined the ground truth for the 10-node synthetic network used in Part I:

```
x1 = (~x2 & ~x6 & ~x5) | (~x2 & ~x6 & x5) | (~x2 & x6 & ~x5) | (x2 & ~
   x6 & ~x5) | (x2 & ~x6 & x5)
x2 = (~x6 & ~x1 & ~x5) | (~x6 & ~x1 & x5) | (~x6 & x1 & ~x5) | (~x6 &
   x1 & x5) | (x6 & ~x1 & x5) | (x6 & x1 & x5)
x3 = 0
x4 = 0
x5 = (~x7 & x5) | (x7 & ~x5)
x6 = (~x6 & ~x10 & ~x4) | (~x6 & ~x10 & x4) | (~x6 & x10 & ~x4) | (~x6
    & x10 & x4) | (x6 & ~x10 & ~x4) | (x6 & x10 & ~x4) | (x6 & x10 &
   x4)
x7 = (x6)
x8 = (~x6 & ~x1 & ~x10) | (~x6 & x1 & ~x10) | (~x6 & x1 & x10) | (x6 &
    ~x1 & ~x10) | (x6 & ~x1 & x10) | (x6 & x1 & ~x10)
x9 = (~x1)
x10 = (~x7 & ~x5) | (~x7 & x5) | (x7 & ~x5)
```

Specification (Size 16):
The following Boolean functions defined the ground truth for the 16-node synthetic network used in Part I:

```
x1 = (x2)
x2 = (~x13 & ~x11 & x5) | (~x13 & x11 & ~x5) | (~x13 & x11 & x5) | (
   x13 & ~x11 & x5) | (x13 & x11 & x5)
x3 = (~x15 & ~x7) | (~x15 & x7)
x4 = (~x15 & ~x4 & ~x13) | (~x15 & x4 & x13) | (x15 & x4 & x13)
x5 = (~x5 & x1)
```

```
x6 = (~x10 & ~x16 & ~x9) | (x10 & ~x16 & ~x9) | (x10 & ~x16 & x9)
x7 = (~x16 & ~x3) | (x16 & x3)
x8 = (x10 & ~x16)
x9 = (~x4 & ~x2) | (~x4 & x2) | (x4 & x2)
x10 = 1
x11 = 0
x12 = 1
x13 = 0
x14 = 1
x15 = 1
x16 = (~x4 & ~x8 & ~x1) | (~x4 & ~x8 & x1) | (~x4 & x8 & ~x1) | (~x4 &
    x8 & x1) | (x4 & ~x8 & ~x1) | (x4 & x8 & ~x1) | (x4 & x8 & x1)
```

B Biological Network Specifications

Specification (Size 6):
The following Boolean functions defined the ground truth for the 6-node biological network used in Part II:

```
x1 = (x3)
x2 = ((~x4 & x6) | x4)
x3 = (~x5 & ~x2)
x4 = (~x5 & ~x1)
x5 = (~x4 & ~x1)
x6 = (~x5 & ~x1)
```